A FINITE MORPHISM OF ALGEBRAIC SPACES IS QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence; the proof is a short reduction to an affine base and to the scheme case, with the two locality lemmas named.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Finiteness and quasi-finiteness may be checked after base change to an affine over the base. Definition definition-integral together with Lemmas lemma-quasi-compact-local and lemma-quasi-finite-local, used at proof lines 9473-9480. Reduces the claim to an affine base Y.
- A finite morphism over an affine base has a scheme as source. used at proof lines 9481-9482. Puts the statement in the setting of schemes.
- The scheme case: a finite morphism is quasi-finite. Morphisms, Lemma morphisms-lemma-finite-quasi-finite, used at proof lines 9482-9483. Is the input applied to the base changed morphism.
- The definition of a finite morphism of algebraic spaces. Definition definition-integral, used at proof lines 9474-9477. Fixes the meaning of finiteness here.

Source reference. Original definition 03ZP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5228–5228. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4676–4676. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 040Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5274–5274. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02NU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3493–3493. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *A finite morphism of algebraic spaces is quasi-finite.*

Proof. Let $f : X \to Y$ be a morphism of algebraic spaces. By Definition 03ZP and Lemmas 03KG and 040Z both properties may be checked after base change to an affine over Y, i.e., we may assume Y affine. If f is finite then X is a scheme. Hence the result follows from the corresponding result for schemes, see Morphisms, Lemma 02NU. □